

3. Schema Design & Data Insertion

Quick Commerce System

3.1 Schema Design Overview

The database schema is designed based on the ER diagram and follows **relational database principles** such as:

- Proper normalization
- Use of **Primary Keys** and **Foreign Keys**
- Appropriate **data types** suitable for SQL*Plus
- Logical attribute grouping

The schema consists of **four tables**:

1. CUSTOMER
 2. PRODUCT
 3. ORDERS
 4. ORDER_ITEM
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3.2 Table Structures (SQL*Plus Compatible)

1. CUSTOMER Table

```
CREATE TABLE customer (  
    customer_id NUMBER PRIMARY KEY,  
    customer_name VARCHAR2(50) NOT NULL,  
    phone_number VARCHAR2(15) UNIQUE,  
    email VARCHAR2(50),  
    address VARCHAR2(100),  
    registration_date DATE  
);
```

2. PRODUCT Table

```
CREATE TABLE product (  
    product_id NUMBER PRIMARY KEY,  
    product_name VARCHAR2(50) NOT NULL,  
    category VARCHAR2(30),  
    price NUMBER(8,2) CHECK (price > 0),  
    stock_quantity NUMBER CHECK (stock_quantity >= 0),  
    availability_status VARCHAR2(15)  
);
```

3. ORDERS Table

```
CREATE TABLE orders (  
    order_id NUMBER PRIMARY KEY,  
    order_date DATE,  
    order_status VARCHAR2(20),  
    total_amount NUMBER(10,2),  
    customer_id NUMBER,  
    CONSTRAINT fk_customer  
    FOREIGN KEY (customer_id)  
    REFERENCES customer(customer_id)  
);
```

4. ORDER_ITEM Table

```
CREATE TABLE order_item (  
    order_item_id NUMBER PRIMARY KEY,  
    quantity NUMBER CHECK (quantity > 0),  
    price_at_order NUMBER(8,2),  
    order_id NUMBER,  
    product_id NUMBER,  
    CONSTRAINT fk_order  
    FOREIGN KEY (order_id)  
    REFERENCES orders(order_id),  
    CONSTRAINT fk_product  
    FOREIGN KEY (product_id)  
    REFERENCES product(product_id)  
);
```

3.3 Data Insertion (Minimum 5 Records per Table)

Insert Data into CUSTOMER

```
INSERT INTO customer VALUES (1, 'Amit Sharma', '9876543210',  
'amit@gmail.com', 'Mumbai', SYSDATE);  
INSERT INTO customer VALUES (2, 'Neha Patil', '9123456780',  
'neha@gmail.com', 'Pune', SYSDATE);  
INSERT INTO customer VALUES (3, 'Rahul Verma', '9988776655',  
'rahul@gmail.com', 'Delhi', SYSDATE);  
INSERT INTO customer VALUES (4, 'Sneha Joshi', '9012345678',  
'sneha@gmail.com', 'Nagpur', SYSDATE);  
INSERT INTO customer VALUES (5, 'Karan Mehta', '9090909090',  
'karan@gmail.com', 'Ahmedabad', SYSDATE);
```

Insert Data into PRODUCT

```
INSERT INTO product VALUES (101, 'Milk', 'Dairy', 50, 100,  
'Available');  
INSERT INTO product VALUES (102, 'Bread', 'Bakery', 40, 60,  
'Available');  
INSERT INTO product VALUES (103, 'Rice', 'Grocery', 70, 200,  
'Available');  
INSERT INTO product VALUES (104, 'Eggs', 'Poultry', 6, 300,  
'Available');
```

```
INSERT INTO product VALUES (105, 'Soap', 'Personal Care', 30, 150, 'Available');
```

Insert Data into ORDERS

```
INSERT INTO orders VALUES (1001, SYSDATE, 'Delivered', 250, 1);  
INSERT INTO orders VALUES (1002, SYSDATE, 'Pending', 180, 2);  
INSERT INTO orders VALUES (1003, SYSDATE, 'Delivered', 320, 3);  
INSERT INTO orders VALUES (1004, SYSDATE, 'Cancelled', 90, 4);  
INSERT INTO orders VALUES (1005, SYSDATE, 'Delivered', 150, 5);
```

Insert Data into ORDER_ITEM

```
INSERT INTO order_item VALUES (1, 2, 50, 1001, 101);  
INSERT INTO order_item VALUES (2, 1, 40, 1002, 102);  
INSERT INTO order_item VALUES (3, 3, 70, 1003, 103);  
INSERT INTO order_item VALUES (4, 6, 6, 1004, 104);  
INSERT INTO order_item VALUES (5, 2, 30, 1005, 105);
```

3.4 Schema Validation

- ✓ All tables have **Primary Keys**
 - ✓ Foreign keys maintain **referential integrity**
 - ✓ Correct data types used for SQL*Plus
 - ✓ **Minimum 5 meaningful records** inserted per table
 - ✓ Constraints ensure data accuracy
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